

Entities and Relationships

IEM 4723 Information Systems Design

Dr. Paritosh Ramanan

School of Industrial Engineering and Management
Oklahoma State University

Fall 2026

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Entities and Attributes

What is an Entity?

- An entity is a class of persons, places, objects, events, or concepts about which we need to capture and store data, and is named using a singular noun.
- Examples of entities include people such as customers, employees, and students; places such as branch offices and campuses; objects such as products and vehicles; events such as orders and reservations; and concepts such as accounts and courses.
- An entity instance is a single occurrence of an entity.

Source: Whitten and Bentley (2007)

Attributes and Identifiers

STUDENT

Name

.Last Name

.First Name

.Middle Initial

Address

.Street Address

.City

.State or Province

.Country

.Postal Code

Phone Number

.Area Code

.Exchange Number

.Number Within Exchange

Date of Birth

Gender

Race

Major

Grade Point Average

- An attribute is a descriptive property or characteristic of an entity, also called an element, property, or field.
- A physical student can have attributes such as hair color and height, and a data entity has data attributes in much the same way.
- An identifier is an attribute, or a group of attributes, that takes on a unique value for each entity instance, and is sometimes called a key.
- A concatenated identifier combines several data elements, such as last name, first name, and phone number, though this approach can grow long and lead to mistakes.
- An identifier is needed to distinguish one instance of an entity from another.

Documenting Entities and Attributes

- Entity relationship diagrams, or ERDs, provide a way to document entities and the attributes that describe them.
- Several styles of ERD exist, and while the notation differs, the underlying concepts are essentially the same.
- The Chen model is named after its originator, Dr. Peter Chen of LSU, and was later extended to show attributes.
- The Information Engineering model, also known as crow's foot notation, is the notation used throughout this course.
- IDEF1X was adopted as a Federal Information Processing Standard by NIST.
- UML notation is used when object-oriented concepts are included in data modeling.

Entities and Attributes Example

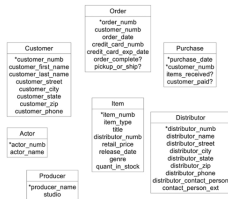


Fig. 4.4 Major entities and their attributes for the Antique Opticals database

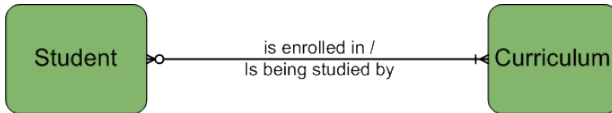
- The IE, IDEF1X, and UML notations list attributes inside the rectangle along with the entity name.
- An entity identifier is marked with an asterisk or the letters PK, such as customer_num (PK).
- Additional entities are still needed to capture the relationships among these entities, which we cover next.

Data Types and Domains

- The data type of an attribute defines the kind of data that can be stored in it, such as CHAR for a fixed-length character string, INT, DECIMAL, DATE, TIME, and BOOLEAN.
- A domain is an expression of the permissible values for an attribute, and the DBMS enforces it through a domain constraint. A domain can be a small user-defined set, such as XS, S, M, L, and XL for sizes.
- Domains are usually not shown in an ER diagram, and are instead captured in a companion document called the data dictionary.
- Choosing the right type or domain for an attribute is critical for database accuracy. A zip code, for example, should be stored as CHAR(10) rather than INT, so that values such as 74078-5018 remain valid.
- Dates should use the DATE domain rather than character strings, since a DBMS may not perform date arithmetic correctly, such as ordering dates or computing an interval, when dates are stored as text.

Data Relationships

What is a Relationship?



- A relationship is a natural business association that exists between instances of one or more entities.
- A relationship may represent an event that links the entities, or simply a logical affinity between instances of the participating entities.
- Cardinality is the minimum and maximum number of instances of one entity that may relate to a single instance of another entity.
- Because every relationship is bidirectional, cardinality must be defined in both directions.

Cardinality Notations

CARDINALITY INTERPRETATION	MINIMUM INSTANCES	MAXIMUM INSTANCES	GRAPHIC NOTATION
Exactly one (one and only one)	1	1	 — or — 
Zero or one	0	1	
One or more	1	many (>1)	
Zero, one, or more	0	many (>1)	
More than one	>1	>1	

Source: Whitten and Bentley (2007)

Relationships Example

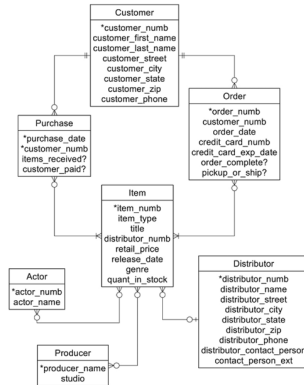
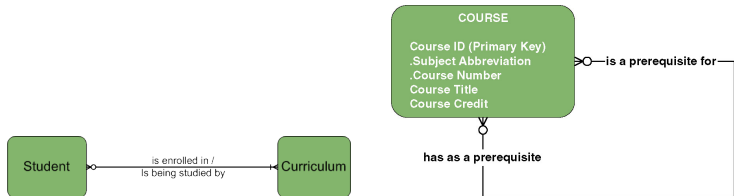


Fig. 4.12 The major entities and the relationships between them in the Antique Opticals database

Degree of a Relationship

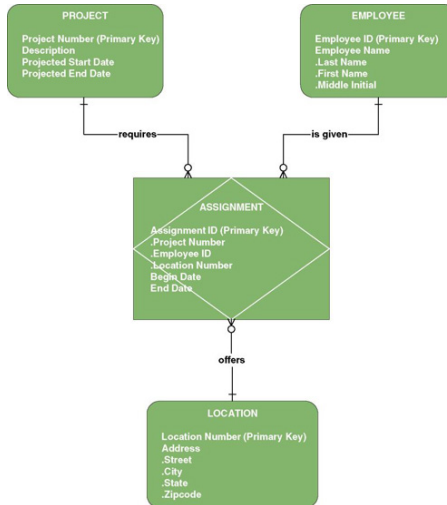
Degree of a Relationship

- The degree of a relationship is the number of entities that participate in it.
- A relationship between two entities is called a binary relationship.
- A relationship between different instances of the same entity is called a recursive, or unary, relationship.



Source: Whitten and Bentley (2007)

Degree of a Relationship, Continued



Keys and Associative Entities

Basic Data Relationships

- There are three basic types of relationships, one-to-one, one-to-many, and many-to-many.
- Relationships stored in a database exist between instances of entities.
- Relationships are captured using foreign keys and associative, or composite, entities.
- One-to-one and one-to-many relationships can be captured through foreign keys.
- Associative or composite entities are needed to capture many-to-many relationships.

Foreign Keys

- A foreign key is the primary key of one entity that is duplicated in another entity to identify instances of a relationship.
- A foreign key always matches the primary key in another entity, though it may or may not be unique itself, and generally is not.
- The entity holding the foreign key is called the child. In a one-to-many relationship, the child is the entity on the many side.
- The entity holding the matching primary key is called the parent. In a one-to-many relationship, the parent is the entity on the one side.

Source: Whitten and Bentley (2007)

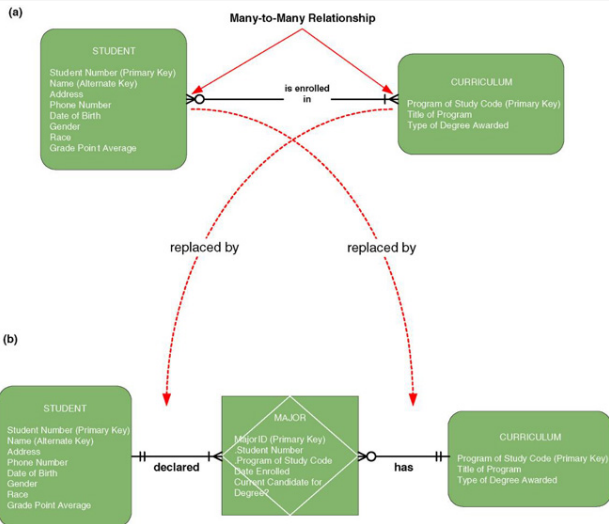
Foreign Keys Example

Student ID	Last Name	First Name	Dorm
2144	Arnold	Betty	Smith
3122	Taylor	John	Jones
3843	Simmons	Lisa	Smith
9844	Macy	Bill	
2837	Leath	Heather	Smith
2293	Wrench	Tim	Jones

Dorm	Director
Smith	Andrea Fernandez
Jones	Daniel Abidjan

- Student ID is the primary key of the Student entity, and Dorm is the primary key of the Dorm entity.
- Dorm in the Student table is a foreign key duplicated from the primary key of the Dorm entity, and it is not unique in the Student entity.

Associative or Composite Entities



Associative Entities, Continued



- Many-to-many relationships can be resolved with an associative entity. Here, the many-to-many relationship between Order and Product is resolved by introducing the Ordered Product associative entity.

Source: Whitten and Bentley (2007)

The Complete ER Diagram

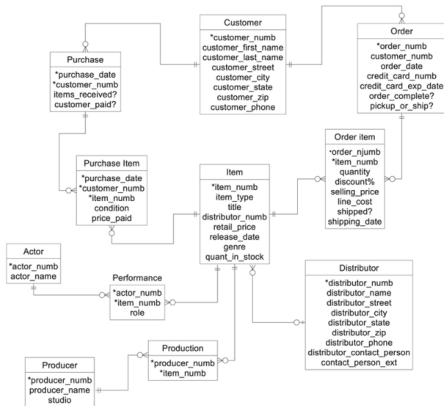


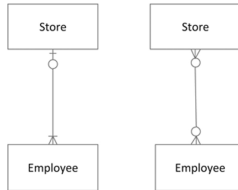
Fig. 4.14 The complete ER diagram for the Antique Opticals database

Business Rules and Model Development

Mandatory and Optional Relationships

- The word zero in phrases such as zero or one and zero, one, or more indicates optionality in a relationship.
- Consider the one-to-many relationship between Customer, the parent entity, and Order, the child entity, in the Antique Opticals ERD. A customer instance can exist without being related to an order instance, so the relationship is optional for the Customer entity, while an order instance must be related to a customer instance, so the relationship is mandatory for the Order entity.
- Consider also the one-to-many relationship between Item, the parent entity, and Distributor, the child entity. This relationship is optional for both entities, since an item can exist without a distributor and a distributor can exist without any item listed in the database.
- Some mandatory or optional constraints are determined by the situation, while others are decided by the business itself.

Relationships and Business Rules



- Harrington (2016) notes that the correct design for a specific business depends on its business rules, and what is correct for one organization may not be correct for another.
- This example models a small establishment with multiple store locations, and asks whether an employee always works at exactly one store or splits time across multiple stores.
- Choosing between these data models depends on how the business actually operates.

Multiple Relationships

- There can be more than one relationship between the same two entities.

Source: Whitten and Bentley (2007)

ER Model Development Stages

- Development of an ER model proceeds through four stages.
- The context data model establishes project scope, and includes only entities and relationships.
- The key-based data model adds key attributes, migrates foreign keys for one-to-many and one-to-one relationships, resolves many-to-many relationships, and adds associative or composite entities.
- The fully-attributed data model adds all remaining descriptive attributes.
- The normalized data model applies normalization, which we cover in an upcoming lecture.

Source: Whitten and Bentley (2007)